

ASSINGMENT – 2

1. Convert the following-
 - a) $(6237.4501)_8 = (\dots\dots\dots)_{10}$
 - b) $(5328.625)_{10} = (\dots\dots\dots)_8$
 - c) $(1101)_{10} = (\dots\dots\dots)_2$
 - d) $(B91D)_{16} = (\dots\dots\dots)_{10}$
 - e) $(3F)_{16} = (\dots\dots\dots)_2$
2. What are universal logic gates? Implement AND and OR gates using universal gates.?
3. What do you understand by Boolean algebra? State and prove
 - a) Demorgan's theorem.
 - b) Commutative law
4. Explain half adder? Implement the full adder using two half adder.
5. Draw the diagram of digital circuit for $F(A,B,C,D) = AB+BC+CD$ using only NAND gate.
6. Minimize the Boolean function using K-map
$$F(A,B,C) = \sum m(0,1,2,4,6)$$
7. Simplify the Boolean expression $F = A'C + A'B + BC + AB'C$ in Sum of product using K-map method.
8. Using Boolean algebra techniques, simplify the following expressions and realize the minimize expression using basic logic gates.
 - a) $AB + \overline{A}C + BC$
 - b) $A + B[AC + (B + \overline{C})D]$
 - c) $A\overline{B}C + B + B\overline{D} + AB\overline{D} + \overline{A}C$
 - d) $ABCD + AB(\overline{CD}) + (\overline{AB})CD$
9. Implement XOR gate using minimum number of NAND Gate.
10. Identify which of the following Boolean expression is incorrect
 - a) $A \text{ XOR } 0 = A'$
 - b) $A \text{ XOR } 1 = A'$

- c) $A \text{ XOR } A = 0$
- d) $A \text{ XOR } A' = 1$
- e) $A + A'B = A + B$
- f) $(A+B)(A+C) = A + BC$